

ALFIDO TECH

Cloud Computing Internship

TASK 2 COMPLETION REPORT

Deploy a Virtual Machine & Install Web Server

Submitted By	Saniya Shaikh
AWS Account	785446018888
Region	US East-1 (N. Virginia)
Instance Name	Saniya-Instance
Instance ID	i-0a80229b314c7718f
Instance Type	t3.micro
Public IP	54.88.158.181
Web Server	Apache2
Security Group	MyWebSG (SSH:22, HTTP:80, HTTPS:443)
Date	April 2026
Status	☒ COMPLETED

Step 1. Launch EC2 Instance (Ubuntu)

An EC2 instance named Saniya-Instance was launched using instance type t3.micro in the US East-1d availability zone. The instance reached a Running state with 3/3 status checks passed. The assigned Public IPv4 address is 54.88.158.181.

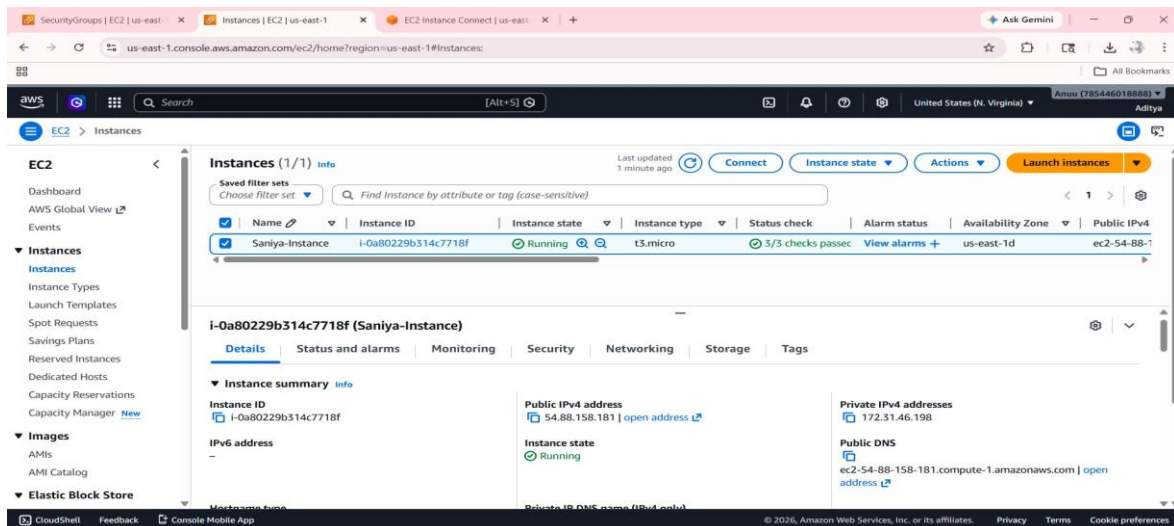


Figure 1 – EC2 instance 'Saniya-Instance' running successfully | Public IP: 54.88.158.181 | Instance ID: i-Oa80229b314c7718f

Step 2. Configure Security Group-Firewall Rules

A security group named MyWebSG was created and attached to the instance. Three inbound rules were configured to allow SSH (port 22), HTTP (port 80), and HTTPS (port 443) traffic over TCP/IPv4, enabling both remote access and public web server access.

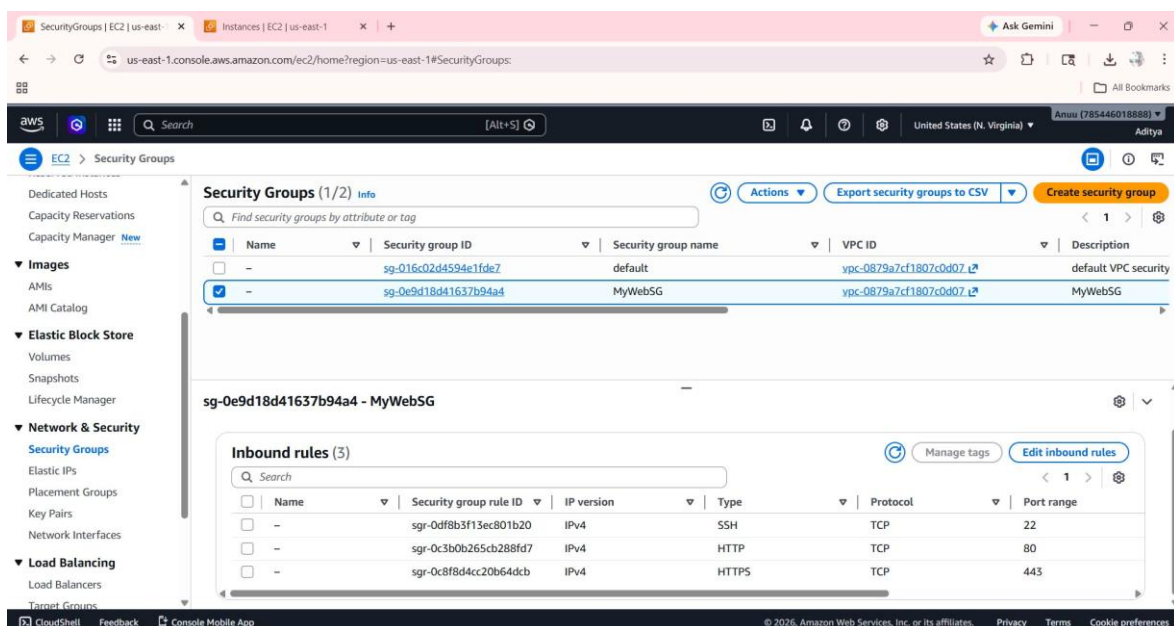
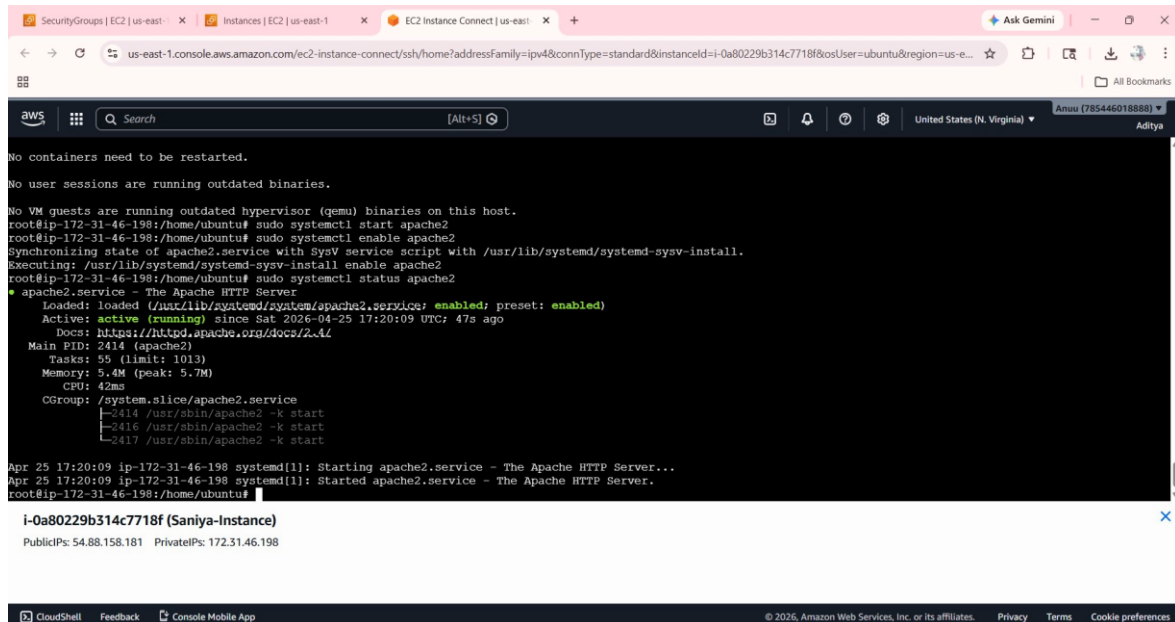


Figure 2 – Security group 'MyWebSG' with inbound rules: SSH (22), HTTP (80), HTTPS (443)

Step 3. Install & Start Apache Web Server

Connected to the instance via EC2 Instance Connect. Apache2 was installed, started, and enabled to launch on reboot. The systemctl status command confirmed Apache is active (running) on instance ip-172-31-46-198.



```
SecurityGroups | EC2 | us-east-1 | Instances | EC2 | us-east-1 | EC2 Instance Connect | us-east-1 | + | Ask Gemini | - | x |
us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0a80229b314c7718f&osUser=ubuntu&region=us-e...
aws | Search | [Alt+S] | United States (N. Virginia) | Anuu (785446018888) | Aditya
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-46-198:/home/ubuntu# sudo systemctl start apache2
root@ip-172-31-46-198:/home/ubuntu# sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
root@ip-172-31-46-198:/home/ubuntu# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-04-25 17:20:09 UTC; 47s ago
     Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2414 (apache2)
    Tasks: 55 (limit: 1013)
  Memory: 5.4M (peak: 5.7M)
     CPU: 42ms
   CGroup: /system.slice/apache2.service
           └─2414 /usr/sbin/apache2 -k start
             └─2416 /usr/sbin/apache2 -k start
             └─2417 /usr/sbin/apache2 -k start
Apr 25 17:20:09 ip-172-31-46-198 systemd[1]: Starting apache2.service - The Apache HTTP Server...
Apr 25 17:20:09 ip-172-31-46-198 systemd[1]: Started apache2.service - The Apache HTTP Server.
root@ip-172-31-46-198:/home/ubuntu#
i-0a80229b314c7718f (Saniya-Instance)
PublicIPs: 54.88.158.181 PrivateIPs: 172.31.46.198
CloudShell | Feedback | Console Mobile App | © 2026, Amazon Web Services, Inc. or its affiliates. | Privacy | Terms | Cookie preferences
```

Figure 3 – Apache2 service showing 'active (running)' status via EC2 Instance Connect terminal

Step 4. Deploy Custom Page — Live Website Verification

A custom 'Hello from Cloud VM' HTML page was deployed to the Apache web root (/var/www/html/index.html). The page is publicly accessible via the EC2 public IP address in the browser, confirming the web server is serving content correctly.

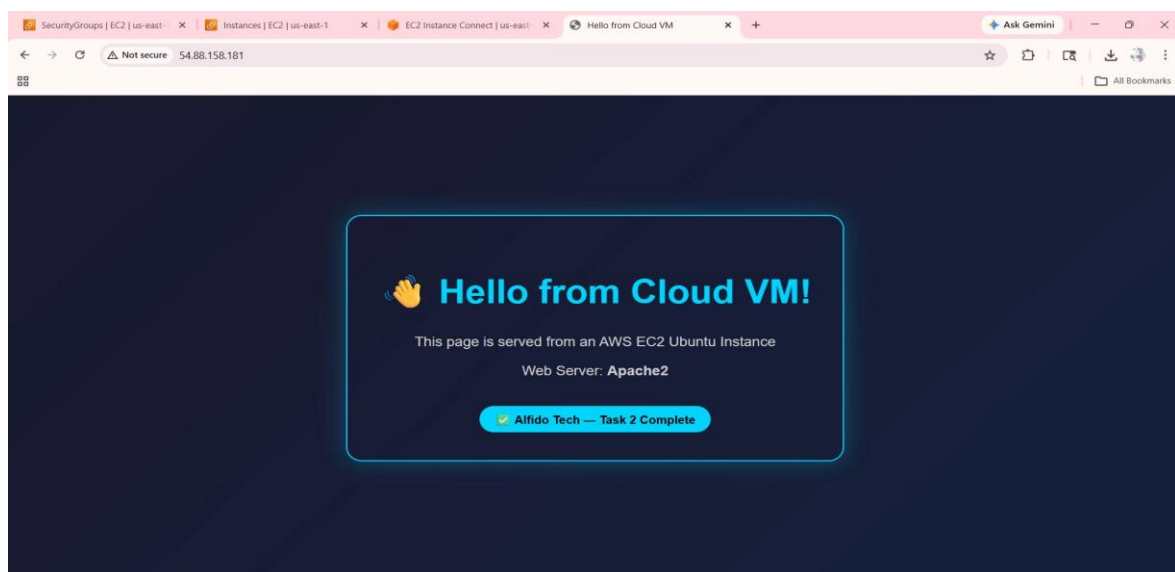


Figure 4 – Live 'Hello from Cloud VM!' page accessible at http://54.88.158.181 via Apache2

Task Completion Summary

Requirement	Action Taken	Status
Create VM Instance (Ubuntu)	Launched t3.micro Ubuntu EC2 (Saniya-Instance)	✓ Done
Configure Firewall (Port 80)	MyWebSG: SSH(22), HTTP(80), HTTPS(443) rules added	✓ Done
Install Apache / nginx	Apache2 installed, enabled & verified active/running	✓ Done
Display "Hello from Cloud VM" page	Custom HTML page deployed to /var/www/html/index.html	✓ Done

Complete Commands Reference

```
# Step 1 - Update system packages
sudo apt update -y

# Step 2 - Install Apache web server
sudo apt install apache2 -y

# Step 3 - Start and enable Apache
sudo systemctl start apache2
sudo systemctl enable apache2

# Step 4 - Verify Apache is running
sudo systemctl status apache2

# Step 5 - Deploy custom HTML page
sudo nano /var/www/html/index.html

# Step 6 - Access the live website
http://54.88.158.181
```

Deliverables

1. Public VM IP with Live Page:

<http://54.88.158.181>

2. Commands Used & Screenshots:

- EC2 instance running — Saniya-Instance (Figure 1)
- Security group MyWebSG with ports 22, 80, 443 open (Figure 2)
- Apache2 active (running) status in terminal (Figure 3)
- Live 'Hello from Cloud VM!' page at <http://54.88.158.181> (Figure 4)
- All commands used documented in Commands Reference section

This report was prepared as part of the Alfido Tech Cloud Computing Internship — Task 2.